

Bowen Zhang

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Summary

- Experience as a Core Java developer, specializing in designing and building enterprise-level applications using **Java/J2EE** and implementing modern **web** technologies.
- Experience in all phases of Software Development Life Cycle (**SDLC**) using **Agile (Scrum)** methodologies, and skilled in building scalable and high-performance solutions in both backend and frontend systems across multiple domains, including Banking and Finance, Online E-Commerce and E-Map.
- Deep comprehension of **Java 8** and **17** features such as **Method Reference**, **Lambda Expression**, **Streams API**, **Optional Class**, **Functional Interface**.
- Extensive hands-on experience in building **RESTful APIs** and web services with backend Spring frameworks, including **Spring MVC** with **Apache Tomcat**, **Spring Boot**, **Spring JPA**, **Spring Cloud** and proficient in **Dependency Injection (IOC)**, and **AOP**.
- Expertise in building responsive user interface with **HTML5**, **CSS3**, **JavaScript/TypeScript** and **Bootstrap**.
- Deep expertise in implementing single page applications (**SPA**) using **React (React 17+ with Hooks and Redux)** and **Angular** frameworks, capable of designing scalable and performant UI components.
- Proficient in using **ORM** model frameworks like **Hibernate**, **Spring Data JPA**, **SQLAlchemy** to do relationship mapping with data entries in backend servers and databases.
- Strong experience in managing **SQL** queries and designing table schema in **RDBMS** systems like **MySQL**, **PostgreSQL** or **Oracle**, and skilled in **NoSQL** databases such as **MongoDB** and **DynamoDB**.
- Experienced in using **Redis** for higher query performance and concurrent operations with distributed lock.
- Strong experience in development with **Microservices** Architecture. Configured the components consisting of **Spring Cloud Config server**, **Load Balancer**, **API Gateway** and **Eureka server**.
- Proficient in **AWS** cloud services, including **RDS**, **ECR/ECS**, **DynamoDB**, **S3**, **CloudFront**, **SNS/SQS**, **Lambda**.
- Experienced in managing the messaging systems for service communication with **Kafka** and **RabbitMQ**.
- Strongly skilled in writing unit tests and integration tests with testing framework of **JUnit** and **Mockito**, and validate the code quality analysis using the library of **PMD** and **SonarQube**.
- Experience with logging from **Spring Actuator**, monitoring and visualizing the data on dashboard by using **Prometheus** and **Grafana**. Managing project tickets and tracking bugs with the tool of **JIRA**.
- Hands on experience in building and packaging projects with tools of **Maven/Gradle**, doing version control with **Git** system, configuring CI/CD pipelines using **Jenkins** and creating containerized apps using **Docker**.

Certificate & Tech Skills

Certification: AWS certified cloud practitioner

Programming Languages: Java, Python, C/C++, Kotlin, JavaScript, TypeScript, HTML/CSS

Tech Frameworks: Spring MVC, Spring Boot, Flask, Spring Data JPA, Hibernate, Spring Security, Spring Cloud, SQLAlchemy, Thymeleaf, FastAPI, Node JS, React, Angular, PyTorch/TensorFlow, LangChain

Databases: MySQL, PostgreSQL, Oracle, Redis, MongoDB, Cassandra/Neo4j, HBase, Elastic Search

Big Data: Hadoop, HDFS, Spark/PySpark, DataFrame

Cloud Technologies: AWS CLI, RDS, Aurora, Athena, DynamoDB, EC2, ECR/ECS, EKS, Lambda, Fargate, SNS/SQS, S3, Route 53, CloudFront, CloudWatch, CloudFormation, CloudTrail, Amplify, IAM, AWS Config

Testing: JUnit, Mockito, Jenkins, Spring Boot Test, SonarQube, PMD, Cucumber

Project Management: Jira, Agile Scrum, Maven, Gradle, NPM, GitLab, Bitbucket

Others: Microservices, Kafka, RabbitMQ, Docker, Kubernetes, GraphQL, Postman, Swagger, Jupyter

Professional Experience

Bank of America

Mar 2024 - May 2025

Senior Java Developer

New York, NY

Project Name: Vehicle Loans System

Description: Developed and maintained a cloud-native Vehicle Loans System with microservice architecture for Bank of America to support end-to-end workflows for auto loan application, approval and repayment processes, ensuring a seamless digital lending experience for individual customers.

Roles & Responsibilities:

- Engaged in **Agile** sprints and ceremonies, and worked closely with cross-functional teams including BAs, product owners, and dev managers to implement business-driven features for the vehicle loans system.
- Followed Behavior-Driven Development (**BDD**) practices using the **Cucumber** framework to define test scenarios, and leveraged **Jira** to assign development tasks, and track completed features and reported issues.
- Developed **RESTful APIs** and delivered scalable, modular backend services using **Spring Boot** within a **microservices** architecture, enabling reliable loan processing and seamless integration across different system components.
- Implemented **Spring AOP** for centralized exception handling to improve system reliability, and applied **transaction management** to ensure data consistency and integrity along the loan processing operations.
- Utilized **Hibernate ORM** to integrate Spring Boot services with **MySQL** and **Oracle DB**, enabling efficient persistence of structured data such as customer profiles, loan application details, vehicle metadata, and approval records.
- Designed and implemented responsive and interactive UI components using **React 17** with functional components and **Redux** for centralized state management, enhancing frontend-backend integration with Spring Boot APIs; optimized **Virtual DOM** rendering reduced page load time by **20%** and improved user responsiveness by **25%**.
- Enabled asynchronous event-driven communication using **Kafka** as the message broker, decoupling the vehicle loans system from external services such as customer account management systems, credit evaluation systems and payment processing systems, and increased the system throughput by **40%**, enabling processing of over **10,000** loan-related events per minute across distributed services.
- Configured key microservices infrastructure components for the system, including **Eureka** for service registry and discovery, Redis-based **rate limiting** in the **API Gateway**, **NGINX** for load balancing, and **Prometheus-Grafana** tech-stack for observability, which reduced the average API response time by **25%** and also decrease the alert response time by **30%**.
- Implemented an alerting mechanism using **AWS SNS** and **SQS** to detect repayment failures or insufficient account balances, triggering automated email and SMS notifications to bank customers for timely awareness.

- Performed unit and integration testing using **JUnit** and **Mockito** to ensure backend service reliability, and leveraged **SonarQube** for continuous code quality analysis, identifying code smells and coverage gaps.
- Used **GitLab** and **Maven** to build and manage project artifacts, and deployed the servers in containerized environments using **Docker** and **Kubernetes** clusters for scalable and consistent delivery.

Macy's

Jan 2023 - Mar 2024

Core Java Developer

New York, NY

Project Name: Buyer Personalization System

Description: Developed a buyer personalization system for Macy's web application to deliver the customized product recommendations, including dynamic homepage feeds and personalized item suggestions based on real-time user behaviors.

Roles & Responsibilities:

- Collaborated with business analysts and data engineers to implement and optimize the buyer personalization system, following **Agile Scrum** methodology and using **Jira** to manage sprint planning, task assignments, and bug tracking throughout the development lifecycle.
- Designed and developed scalable and high-performance **RESTful APIs** using the **Spring Boot** framework, following Behavior-Driven Development (**BDD**) practices with tool of **Cucumber** to ensure feature alignment with business requirements.
- Implemented dynamic and modular UI components using the **Angular** framework with **TypeScript** to render the Personalized Shop and Discover sections, featuring carousel displays of recommended items across multiple categories such as clothing, beauty, jewelry, furniture, and electronics; utilized **Angular Router, DI** with **NPM** for efficient management of component dependencies and navigation.
- Developed **microservices** for user profiling, preference modeling, and recommendation delivery, integrating **Kafka** as the message broker to enable asynchronous data streaming and decoupled service communication.
- Utilized **PostgreSQL** instances on **AWS RDS** from AWS cloud services to store structured data such as product metadata, user behavior logs, and profile attributes, while leveraging **AWS DynamoDB** to manage flexible data including recommendation mappings, session state, and product image collections.
- Applied **Redis** server to cache frequently accessed recommendation results and hot product data, reducing redundant computation and improving response time for users.
- Integrated **ElasticSearch** to enable hybrid search combined with personalized ranking, retrieving products based on user queries and prioritizing items aligned with individual browsing behaviors and preferences.
- Designed and executed unit tests with **JUnit** and integration tests using **Mockito**, ensuring code correctness, service isolation, and API contract stability.
- Conducted code quality analysis using **SonarQube** for each sprint's new feature implementations, identifying code smells, measuring code coverage, ensuring the maintainability and reliability across backend and frontend modules.
- Built **CI/CD** pipelines with **Jenkins** to automate the deployment of containerized apps, and integrated **Spring Boot Actuator** with **Prometheus** and **Grafana** to monitor service health metrics and visualize real-time logs.

Airbnb

Jul 2020 - Dec 2022

Java Backend Developer

Beijing, China

Project Name: Online Homestay Rental Platform

Description: Built and maintained a full-stack homestay rental platform at Airbnb, supporting end-to-end booking workflows including categorized listings, search, reservation, booking, and payment processing.

Roles & Responsibilities:

- Participated in all phases of the software development lifecycle, including requirement analysis, system design, project development, API documentation and code testing with **Agile/Scrum** methodologies.
- Implemented function-driven **RESTful APIs**, and maintained high-performance and scalable web services in microservice architecture using **Spring Boot**, **Spring Data JPA**, **PostgreSQL** and **Redis**.
- Integrated **OAuth 2.0** authentication framework using **Keycloak** as the authorization server to manage secure user authentication and role-based authorization within the homestay booking system.
- Utilized **AWS RDS** to store structured data such as user profiles and property listings, and integrated **AWS DynamoDB** to manage unstructured data including listing metadata, images, and multimedia content.
- Managed and deployed containerized application servers on **AWS** using **ECR/ECS** and **Amazon EKS** for scalable orchestration, leveraged **S3** for static content storage, and integrated **CloudFront** as a **CDN** to reduce latency and improve user experience.
- Designed and implemented responsive UI components for the web application using **Angular** and **Angular CLI**, leveraging **Node.js** and **NPM** for efficient dependency management and development.
- Utilized **Apache Kafka** to handle real-time event streaming in the homestay booking platform, enabling high scalability and fault-tolerant processing during peak traffic periods.
- Integrated **Redis** to cache recently viewed and trending homestay listings, and leveraged **Elasticsearch's** geo-based search capabilities to deliver location-aware property recommendations to users.
- Practiced Test-Driven Development (**TDD**) by writing unit tests with JUnit and performing backend integration testing using **Spring Boot Test**; applied **PMD** for code quality analysis to detect code style violations, potential bugs, and unused variables or blocks.

ByteDance

Java Backend Engineer

Aug 2018-Jul 2020

Chengdu, China

Project Name: E-commerce User Management Portal

Description: Developed the User Management Portal for TikTok Shop, a centralized system that manages both merchant and customer accounts, enabling role-based access control, authentication, and personalized dashboard features based on user credentials.

Roles & Responsibilities:

- Developed efficient and highly scalable **REST** based web services to build E-commerce solutions using the backend framework of **Spring MVC**, and integrated external partner APIs to scale up the business.
- Participated in **AgileScrum** development lifecycle with daily stand-ups and sprint planning, and applied Test-Driven Development (**TDD**) practices to collaboratively implement new features and maintain the system.
- Utilized **Hibernate ORM** with **MySQL** to store structured data such as user credentials, roles and account settings, and imported **MongoDB** to handle unstructured data like the user activity logs and related images.
- Integrated template engine from **Thymeleaf** to build connection with frontend and backend system. Designed and optimized interactive UI interfaces with web technologies of **HTML5**, **CSS3**, **JavaScript**.
- Secured RESTful APIs using **Spring Security** with **JWT** tokens and session-based cookies to implement robust **authentication** and role-based **authorization** for both merchants and customers.

- Used **RabbitMQ** with point-to-point model as the distributed message queue to handle high-concurrency tasks, and ensured the asynchronous communication between different microservice servers.
- Wrote unit tests and integration tests using tools of **JUnit/Mockito** and do the API testing by **Postman**.
- Containerized applications using **Docker** and managed distributed deployment with **Kubernetes** cluster.

Baidu

Backend Software Developer

Sep 2016 - Aug 2018

Chengdu, China

Project Name: Data Processing & Visualization System

Description: Developed a data preprocessing pipeline project for Baidu Maps' algorithm models using big data technologies, with backend automation on remote Linux servers and frontend integration into web systems.

Roles & Responsibilities:

- Implemented projects using **Agile** methodology, actively participated in weekly meetings, sprint showcases and retrospectives.
- Developed backend web services for display dashboard with microservices architecture using the Python **Flask** framework.
- Developed efficient RESTful API solutions by building Java applications leveraging core features such as **Collections**, **multithreaded pools**, **lambda expressions**, and **Stream API**; tested endpoint functionality with **Postman** and utilized **Swagger** for automatic API documentation.
- Built modular frontend UI components using **React 16** with functional components and **Ant Design** library, integrating seamlessly with **REST APIs** and **microservices** for real-time data display in dashboard interface.
- Displayed the processing data and final results on a web dashboard and visualized data with support of **Baidu Echarts** library.
- Implemented microservice architecture with **API Gateway**, **Eureka** server for service registry and discovery, and **Nginx** server for load balancing to support scalable and modular data processing.
- Used the **Hadoop** platform to store large-scale datasets on distributed file system of **HDFS**, and employed **Hadoop Streaming** library with a map-reduce structure to process data in batch at high concurrency.
- Integrated **Kafka** as the message broker to enable real-time data streaming and complement the batch processing workflows.
- Managed and reviewed project code using **GitLab**, and integrated **GitHub Actions** as the **CI/CD pipeline** to automate testing and packaging processes.

Education

New York University

New York, US

- *Master of Science in Computer Engineering 2024*

Southwest Jiaotong University

Chengdu, China

- *Bachelor of Science in Computer Science & Technology 2019*